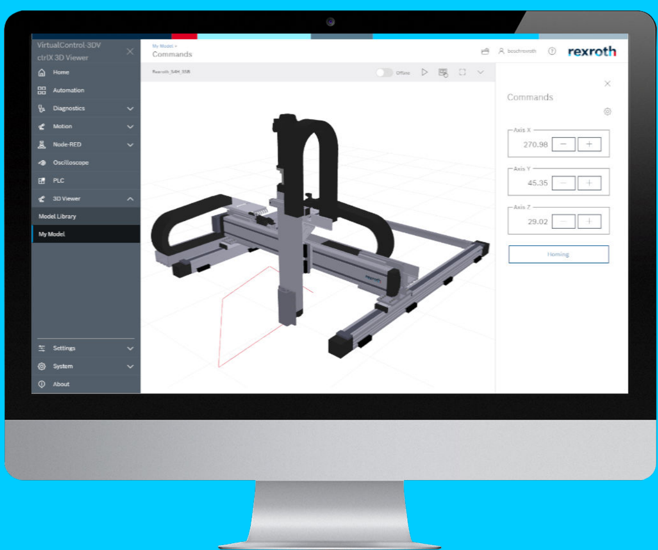
Release Notes

3D Viewer App

Browser-based 3D Kinematic Simulation

Version 4.6.1



Edition 02

Disclaimer

The specified data only serves to describe the product. As our products are constantly being further developed, no statements concerning a certain condition or suitability for a certain application can be derived from our information. The information given does not release the user from the obligation of own judgment and verification. Please note that our products are subject to a natural process of wear and aging.

DC-AE/PAX (TaDo/PiaSt)

1b2ae187738109dd0a6800264ccee997, 2, en_US

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1 About this documentation

Editions of this documentation

Edition	Release date	Note
01	2025-11	3D Viewer App Version 4.4.0
02	2026-03	3D Viewer App Version 4.6.1

Further information

- See ➦3D Viewer App, Application Manual

2 Important information

2.1 Product information

For the latest information on our products, go to ➦ https://www.boschrexroth.com.

For current security advisories, please refer to ➦ https://psirt.bosch.com/security-advisories/

3 Version 4.6.1

3.1 New functions

3.1.1 Feature 1132223: Maintenance and improvements to the 3D Viewer Widget

Description

- Upgrade to Angular20 and Core24
- 3DViewer widget improvements
- User interface improvements

3.2 Resolved defects and modifications

3.2.1 Bug 1159420: Menu buttons overlap with small window sizes such as IFrame and Widget

Description

The menu buttons for the settings overlap each other in small window sizes such as IFrame and Widget. In addition, the usability via a horizontal scrollbar was unfavorable.

Troubleshooting

The menu is displayed in multiple lines if required.

3.2.2 Bug 1169015: The lossless subscription does not work

Description

The real-time subscription did not work. When the real-time subscription was activated, only a conventional subscription took effect on the nodes of the Data Layer that are intended for real-time data. As a result, too few interpolation points were transferred to the 3D Viewer and contour deviations were detected in the tool path.

Troubleshooting

The error correction requires the use of weblib.automationcore version 11.0 or higher.

3.2.3 Bug 1180546: Incorrect display of WCS-based safety and workspaces

Description

An incorrectly calculated transformation was used to display the WCS-based safety and work areas, which resulted in an incorrect position and orientation of the graphic.

In addition, there was an error when refreshing the graphics for teach points and areas during the Motion state transition from "Configuration" to "Running".

3.2.4 Bug 1187818: Graphic implementation of safety/working areas, tool length corrections and changes to the PCS at the wrong time

Description

Previously, activations/deactivations of safety/working areas, tool length corrections and the PCS took effect graphically as soon as they were included in the corresponding Data Layer node for prepared commands. Among other things, this resulted in a distorted tool path if a length compensation change was prepared during the execution of a previous command.

Troubleshooting

For the graphic implementation of safety/working areas, tool length corrections and the PCS, the content of the corresponding Data Layer node that reflects the active commands is decisive.

3.2.5 Bug 1188775: Incorrect calculation of the orientation of the MCS with effect on the graphic for teach points and areas

Description

During the axis transformation for a 6-axis kinematic with spherical hand, the orientation of the machine coordinate system, which forms the basis for the display of ranges and teach points, was calculated incorrectly if the axis direction for the assigned joints Rot1 or Rot2 is set to "Negative".

4 Version 4.4.0

4.1 New functions

4.1.1 Feature 777649: Considering the tool length compensation in the tool path

Description

The resulting tool path line can take into account an effective tool length compensation. The drawing point is then offset by the correction vector in relation to the zero point at the tool node.

4.1.2 Feature 945240: Extending the display of work and protection zones to PCS

Description

Work and protection zones defined for a product coordinate system (PCS) can be displayed graphically.

4.1.3 Feature 1001439: Workpieces predefined in the machine model

Description

In the machine model (XML file), STL files can be specified as geometry resources for a workpiece in case of workpiece nodes (StockNode). The STL files were ignored in previous versions.

These predefined workpieces are visible immediately after importing the machine model with the attributes from the model file and can subsequently to be changed or deleted. It is only possible to influence their visibility. The predefined workpieces are assigned the name of the workpiece node.

4.1.4 Feature 1008627: Support for parallel kinematics

Description

The 3D Viewer supports parallel kinematics in connection with crlX Motion, but creating the machine model is very challenging. It is therefore possible to create simplified models via a program, which can be refined or expanded later if necessary.

In addition to existing models, such models can also be created for the following transformation types:

- 3-axis delta (linear axes)
- 3-axis adabot
- 3-axis twisting crane

If required, refer to your Bosch Rexroth customer service.

4.1.5 Feature 1055489: Extending the display in the kinematic environment (Motion)

Description

The following function extensions have been implemented for the graphical representation of processes controlled by kinematics:

- Defined work and safety areas can be displayed in the world coordinate system (WCS).
- The optional display of the flange coordinate system (FCS) facilitates to visually assess the current orientation of the tool head and the effective direction of the tool length compensation.

4.1.6 Feature 1078521: Displaying ACS-based teach points

Description

The display of teach points has been expanded to include the points defined in the axis coordinate system.

4.2 Resolved defects and modifications

4.2.1 Bug 1045324: Memory requirement for the preview image

Description

The memory requirement for the preview graphic, which is displayed on the tiles of the model library, was too high and in need of optimization.

Bugfixing

Among other things, the images are saved in a file format with higher data compression.

Caution:

The use of an alternative file format does not guarantee downward compatibility. Projects that have been imported with a 3D Viewer version 4.2 or higher do not show a preview image in a 3D Viewer version 3.6 or lower. Remedy by deleting and re-importing the project.

4.2.2 Bug 1106971: Unsaved settings for the toolpath

Description

If a model has more than one tool node, the settings for the tool path (e.g. the drawing "Always on Top") for the second and each subsequent tool node were not saved in the ctrlX.

5 Service and support

Our worldwide service network provides an optimized and efficient support. Our experts provide you with advice and assistance. You can contact us **24/7**.

Service Germany

Our technology-oriented Competence Center in Lohr, Germany, is responsible for all your service-related queries for electric drive and controls.

Contact the **Service Hotline** and **Service Helpdesk** under:

Phone:	+49 9352 40 5060
Fax:	+49 9352 18 4941
Email:	➦service.svc@boschrexroth.de
Internet:	➦http://www.boschrexroth.com

Additional information on service, repair (e.g. delivery addresses) and training can be found on our internet sites.

Service worldwide

Outside Germany, please contact your local service office first. For hotline numbers, refer to the sales office addresses on the internet.

Preparing information

To be able to help you more quickly and efficiently, please have the following information ready:

- Detailed description of malfunction and circumstances
- Type plate specifications of the affected products, in particular type codes and serial numbers
- Your contact data (phone and fax number as well as your e-mail address)

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